

Qn	Suggested Solutions
1(i)	$n = 2,$ $\text{RHS} = \sum_{r=0}^1 \left(-\frac{2}{3}\right)^r + \frac{4}{3} \sum_{r=0}^0 \left(-\frac{2}{3}\right)^r$ $= \left(1 - \frac{2}{3}\right) + \frac{4}{3}$ $= \frac{5}{3}$ $\text{LHS} = x_2$ $= \frac{1}{3}x_1 + \frac{2}{3}x_0$ $= \frac{1}{3}(1) + \frac{2}{3}(2) = \frac{5}{3}$ $n = 3,$ $\text{RHS} = \sum_{r=0}^2 \left(-\frac{2}{3}\right)^r + \frac{4}{3} \sum_{r=0}^1 \left(-\frac{2}{3}\right)^r$ $= \left(1 - \frac{2}{3} + \frac{4}{9}\right) + \frac{4}{3} \left(1 - \frac{2}{3}\right)$ $= \frac{11}{9}$ $\text{LHS} = x_3$ $= \frac{1}{3}x_2 + \frac{2}{3}x_1$ $= \frac{1}{3}\left(\frac{5}{3}\right) + \frac{2}{3}(1)$ $= \frac{11}{9}$
1(ii)	Let $P(n)$ be the statement (*). By (i), $P(2), P(3)$ is true for $n = 2, 3$. Assume $P(k), P(k+1)$ is true for some $k \in \mathbb{N}^+, k \geq 2$. $x_k = \sum_{r=0}^{k-1} \left(-\frac{2}{3}\right)^r + \frac{4}{3} \sum_{r=0}^{k-2} \left(-\frac{2}{3}\right)^r, \quad x_{k+1} = \sum_{r=0}^k \left(-\frac{2}{3}\right)^r + \frac{4}{3} \sum_{r=0}^{k-1} \left(-\frac{2}{3}\right)^r$ Consider $P(k+2)$, we have $x_{k+2} = \frac{1}{3}x_{k+1} + \frac{2}{3}x_k$.

	$ \begin{aligned} x_{k+2} &= \frac{1}{3} \left\{ \sum_{r=0}^k \left(-\frac{2}{3} \right)^r + \frac{4}{3} \sum_{r=0}^{k-1} \left(-\frac{2}{3} \right)^r \right\} \\ &+ \frac{2}{3} \left\{ \sum_{r=0}^{k-1} \left(-\frac{2}{3} \right)^r + \frac{4}{3} \sum_{r=0}^{k-2} \left(-\frac{2}{3} \right)^r \right\} \\ &= \frac{1}{3} \left\{ \sum_{r=0}^{k-1} \left(-\frac{2}{3} \right)^r + \left(-\frac{2}{3} \right)^k + \frac{4}{3} \left[\sum_{r=0}^{k-2} \left(-\frac{2}{3} \right)^r + \left(-\frac{2}{3} \right)^{k-1} \right] \right\} \\ &+ \frac{2}{3} \left\{ \sum_{r=0}^{k-1} \left(-\frac{2}{3} \right)^r + \frac{4}{3} \sum_{r=0}^{k-2} \left(-\frac{2}{3} \right)^r \right\} \\ &= \sum_{r=0}^{k-1} \left(-\frac{2}{3} \right)^r + \frac{4}{3} \sum_{r=0}^{k-2} \left(-\frac{2}{3} \right)^r + \frac{1}{3} \left(-\frac{2}{3} \right)^k + \frac{4}{9} \left(-\frac{2}{3} \right)^{k-1} \\ &= \left\{ \sum_{r=0}^{k-1} \left(-\frac{2}{3} \right)^r + \frac{1}{3} \left(-\frac{2}{3} \right)^k \right\} + \frac{4}{3} \left\{ \sum_{r=0}^{k-2} \left(-\frac{2}{3} \right)^r + \frac{1}{3} \left(-\frac{2}{3} \right)^{k-1} \right\} \\ &= \left\{ \sum_{r=0}^{k-1} \left(-\frac{2}{3} \right)^r + \left(1 - \frac{2}{3} \right) \left(-\frac{2}{3} \right)^k \right\} \\ &+ \frac{4}{3} \left\{ \sum_{r=0}^{k-2} \left(-\frac{2}{3} \right)^r + \left(1 - \frac{2}{3} \right) \left(-\frac{2}{3} \right)^{k-1} \right\} \\ &= \left\{ \sum_{r=0}^{k-1} \left(-\frac{2}{3} \right)^r + \left(-\frac{2}{3} \right)^k \left(-\frac{2}{3} \right)^{k+1} \right\} \\ &+ \frac{4}{3} \left\{ \sum_{r=0}^{k-2} \left(-\frac{2}{3} \right)^r + \left(-\frac{2}{3} \right)^{k-1} \left(-\frac{2}{3} \right)^k \right\} \\ &= \sum_{r=0}^{k+1} \left(-\frac{2}{3} \right)^r + \frac{4}{3} \sum_{r=0}^k \left(-\frac{2}{3} \right)^r \\ &P(k), P(k+1) \Rightarrow P(k+2) \text{ is true.} \\ &\text{By Mathematical Induction, } P(n) \text{ is true for } n \geq 2, n \in \mathbb{N}^+. \end{aligned} $
1(iii)	Now since $\left -\frac{2}{3} \right < 1$, $ \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \left(-\frac{2}{3} \right)^r = \lim_{n \rightarrow \infty} \frac{1 - \left(-\frac{2}{3} \right)^n}{1 - \left(-\frac{2}{3} \right)} = \frac{3}{5}, \text{ and similarly } \lim_{n \rightarrow \infty} \sum_{r=0}^{n-2} \left(-\frac{2}{3} \right)^r = \frac{3}{5}. $ $ \therefore \lim_{n \rightarrow \infty} x_n = \frac{3}{5} + \frac{4}{3} \left(\frac{3}{5} \right) = \frac{7}{5} $

2a(i)	Suppose, on the contrary, that $\exists p, q \in \mathbb{R}$ such that $p < q$ and $f^{-1}(p) \leq f^{-1}(q)$
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	(increasing). Since f is strictly decreasing and bijective, $f[f^{-1}(p)] \geq f[f^{-1}(q)]$. But $p \geq q$ which is a contradiction. $\therefore f^{-1}$ is strictly decreasing.
2a(ii)	From (i), f^{-1} is strictly decreasing, and given $a_1 \leq a_k \leq a_n$ where $1 \leq k \leq n$. It implies that $f^{-1}(a_n) \leq f^{-1}(a_k) \leq f^{-1}(a_1)$. Thus, $\sum_{k=1}^n f^{-1}(a_n) \leq \sum_{k=1}^n f^{-1}(a_k) \leq \sum_{k=1}^n f^{-1}(a_1)$ $nf^{-1}(a_n) \leq \sum_{k=1}^n f^{-1}(a_k) \leq nf^{-1}(a_1)$ $f^{-1}(a_n) \leq \frac{1}{n} \sum_{k=1}^n f^{-1}(a_k) \leq f^{-1}(a_1)$
2a(iii)	(1) $\forall p, q \in \mathbb{R}$ $g(p) = g(q)$ $\frac{1}{p^3} = \frac{1}{q^3}$ $\frac{1}{q^3} = \frac{1}{p^3}$ $q = p$ g is injective. (2) $\forall p \in \mathbb{R} \setminus \{0\}, \exists \frac{1}{x^3} \in \mathbb{R} \setminus \{0\}$ Now, $g\left(\frac{1}{x^3}\right) = \frac{1}{\left(\frac{1}{x^3}\right)^3}$ $= \frac{1}{\frac{1}{x^3}} = x$ g is surjective. (1) and (2) implies g is bijective. $\forall x \neq 0$ $g'(x) = -\frac{1}{3x^4} < 0$ $\therefore g$ is strictly decreasing. Also, $g^{-1}(x) = \frac{1}{x^3}$ and $1 \leq k \leq n$, From (ii),

	$g^{-1}(n) \leq \frac{1}{n} \sum_{k=1}^n g^{-1}(k) \leq g^{-1}(1)$ $\frac{1}{n^3} \leq \frac{1}{n} \sum_{k=1}^n \frac{1}{k^3} \leq 1$ $\therefore \frac{1}{n^3} \leq \frac{1}{n} \left(1 + \frac{1}{2^3} + \frac{1}{3^3} + \dots + \frac{1}{n^3} \right) \leq 1$
2b(i)	$\sum_{r=1}^n \frac{1}{r(r+1)} = \sum_{r=1}^n \left[\frac{1}{r} - \frac{1}{r+1} \right]$ $= 1 - \frac{1}{2}$ $+ \frac{1}{2} - \frac{1}{3}$ $+ \dots$ $+ \frac{1}{n} - \frac{1}{n+1}$ $= 1 - \frac{1}{n+1}$ $= \frac{n}{n+1}$
2b(ii)	Apply $AM \geq GM$ to n numbers and by considering $\frac{1}{1 \cdot 2}, \frac{1}{2 \cdot 3}, \dots, \frac{1}{n(n+1)}$, $\frac{1}{n} \sum_{r=1}^n \frac{1}{r(r+1)} \geq \sqrt[n]{\frac{1}{1 \cdot 2} \cdot \frac{1}{2 \cdot 3} \cdot \dots \cdot \frac{1}{n(n+1)}}$ $= \sqrt[n]{\frac{1}{(n!)^2 (n+1)}}$ $\left[\frac{1}{n} \left(\frac{n}{n+1} \right) \right]^n \geq \frac{1}{(n!)^2 (n+1)}$ $\frac{1}{(n+1)^{n-1}} \geq \frac{1}{(n!)^2}$ $(n!)^2 \geq (n+1)^{n-1}$ $n! \geq (n+1)^{\frac{n-1}{2}} \quad (\text{Shown})$

3(i)	$\tan\left[\frac{\pi}{2} - \tan^{-1}(\sqrt{2} + 1)\right] = \frac{1}{\tan\left[\tan^{-1}(\sqrt{2} + 1)\right]}$ $= \frac{1}{\sqrt{2} + 1}$ $= \frac{1}{\sqrt{2} + 1} \left(\frac{\sqrt{2} - 1}{\sqrt{2} - 1} \right)$ $= \sqrt{2} - 1$ Thus, $\frac{\pi}{2} - \tan^{-1}(\sqrt{2} + 1) = \tan^{-1}(\sqrt{2} - 1)$
3(ii)	Let $\frac{1}{1+t^4} = \frac{At+B}{t^2-\sqrt{2}t+1} + \frac{Ct+D}{t^2+\sqrt{2}t+1}$. $1 = (At+B)(t^2+\sqrt{2}t+1) + (Ct+D)(t^2-\sqrt{2}t+1)$ We have $A+C=0$ $\sqrt{2}A+B-\sqrt{2}C+D=0$ $A+\sqrt{2}B+C-\sqrt{2}D=0$ $B+D=1$ Solving the above equations, we have $A = -\frac{1}{2\sqrt{2}}, B = \frac{1}{2}, C = \frac{1}{2\sqrt{2}}, D = \frac{1}{2}.$ Therefore, $\frac{1}{1+t^4} = \frac{1}{2\sqrt{2}} \left(\frac{-t+2}{t^2-\sqrt{2}t+1} \right) + \frac{1}{2\sqrt{2}} \left(\frac{t+\sqrt{2}}{t^2+\sqrt{2}t+1} \right).$

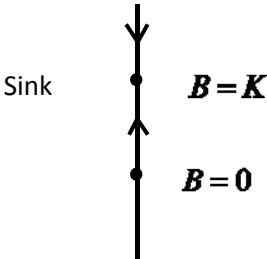
	$ \begin{aligned} \int_0^1 \frac{1}{1+t^4} dt &= \frac{1}{2\sqrt{2}} \int_0^1 \frac{-t+\sqrt{2}}{t^2-\sqrt{2}t+1} dt + \frac{1}{2\sqrt{2}} \int_0^1 \frac{t+\sqrt{2}}{t^2+\sqrt{2}t+1} dt \\ &= -\frac{1}{4\sqrt{2}} \int_0^1 \frac{2t-\sqrt{2}-\sqrt{2}}{t^2-\sqrt{2}t+1} dt + \frac{1}{4\sqrt{2}} \int_0^1 \frac{2t+\sqrt{2}+\sqrt{2}}{t^2+\sqrt{2}t+1} dt \\ &= -\frac{1}{4\sqrt{2}} \int_0^1 \frac{2t-\sqrt{2}}{t^2-\sqrt{2}t+1} dt + \frac{1}{4} \int_0^1 \frac{1}{\left(t-\frac{\sqrt{2}}{2}\right)^2 + \frac{1}{2}} dt \\ &\quad + \frac{1}{4\sqrt{2}} \int_0^1 \frac{2t+\sqrt{2}}{t^2+\sqrt{2}t+1} dt + \frac{1}{4} \int_0^1 \frac{1}{\left(t^2+\frac{\sqrt{2}}{2}\right)^2 + \frac{1}{2}} dt \\ &= -\frac{1}{4\sqrt{2}} \left[\ln(t^2-\sqrt{2}t+1) \right]_0^1 + \frac{1}{2\sqrt{2}} \left[\tan^{-1} \left(\frac{t-\frac{\sqrt{2}}{2}}{\frac{1}{\sqrt{2}}} \right) \right]_0^1 \\ &\quad + \frac{1}{4\sqrt{2}} \left[\ln(t^2+\sqrt{2}t+1) \right]_0^1 + \frac{1}{2\sqrt{2}} \left[\tan^{-1} \left(\frac{t+\frac{\sqrt{2}}{2}}{\frac{1}{\sqrt{2}}} \right) \right]_0^1 \\ &= -\frac{1}{4\sqrt{2}} \ln(2-\sqrt{2}) + \frac{1}{2\sqrt{2}} \tan^{-1}(\sqrt{2}-1) - \frac{1}{2\sqrt{2}} \left(-\frac{\pi}{4} \right) \\ &\quad + \frac{1}{4\sqrt{2}} \ln(2+\sqrt{2}) + \frac{1}{2\sqrt{2}} \tan^{-1}(\sqrt{2}+1) - \frac{1}{2\sqrt{2}} \left(\frac{\pi}{4} \right) \\ &= \frac{1}{4\sqrt{2}} \left[\ln(2+\sqrt{2}) - \ln(2-\sqrt{2}) \right] \\ &\quad + \frac{1}{2\sqrt{2}} \left[\tan^{-1}(\sqrt{2}-1) + \tan^{-1}(\sqrt{2}+1) \right] \\ &= \frac{1}{4\sqrt{2}} \left[\ln(2+\sqrt{2}) - \ln(2-\sqrt{2}) \right] + \frac{1}{2\sqrt{2}} \left(\frac{\pi}{2} \right) \\ &= \frac{1}{4\sqrt{2}} \left[\ln(2+\sqrt{2}) - \ln(2-\sqrt{2}) + \pi \right] \end{aligned} $
3(iii)(a)	For all $t \in [0, x]$,

	$\frac{1}{1+t^4} \leq 1$ $\frac{t^{4n}}{1+t^4} \leq t^{4n}$ $\int_0^x \frac{t^{4n}}{1+t^{4n}} dt \leq \int_0^x t^{4n} dt$ $= \left[\frac{t^{4n+1}}{4n+1} \right]_0^x$ $= \frac{x^{4n+1}}{4n+1}$
3(iii)(b)	For all $t \in [0, x]$, $\frac{1}{1+t^4} \geq 0$. Therefore, from (iii)(a) $0 \leq \int_0^x \frac{t^{4n}}{1+t^4} dt \leq \frac{x^{4n+1}}{4n+1}$ Since $0 \leq x \leq 1$, $\lim_{n \rightarrow \infty} \frac{x^{4n+1}}{4n+1} = 0.$ By Squeeze Theorem, $\lim_{n \rightarrow \infty} \int_0^x \frac{t^{4n}}{1+t^4} dt = 0$.
4(i) (a)	This is equivalent of distributing m identical balls into 4 distinct boxes, so The number of ways = ${}^{m+3}C_3$
4(i)(b)	When m is odd, the number of ways = 0 When m is even, there are $m/2$ even and odd numbers respectively. For even numbers, it can only be 0 or 2, so the number of ways is equivalent of finding the number of way to distribute $m/2$ identical balls into 2 distinct boxes, i.e. $\frac{m}{2} + 1$. It is similar for odd numbers. Hence, the total number of ways = $\left(\frac{m}{2} + 1 \right)^2$
4(ii)	Let a_n be the legitimate number of codewords with length n, For any codeword of length n, we can add another number at the end of the codeword to form a codeword of length $n+1$, If the codeword of length n is a legitimate codeword, we can add 1, 2 or 3 as the last number to ensure that the new codeword of length $n+1$ is still legitimate. If the codeword of length n is not a legitimate codeword, we can add 0 as the last number to ensure that the new codeword of length $n+1$ is now legitimate. Hence, $a_{n+1} = 3a_n + (4^n - a_n) = 2a_n + 4^n$ And

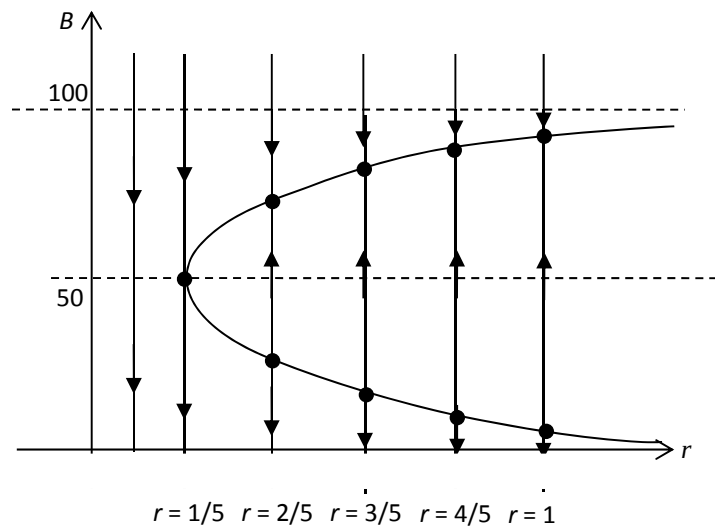
	$a_1 = 3$ $a_2 = 2(3) + 4 = 10$ $a_3 = 2(10) + 4^2 = 36$ $a_4 = 2(36) + 4^3 = 136$
4(iii)	$(p - q)^m$
4(iv)	Let A_i be the event that the number in the i^{th} position of the codeword is chosen from $\{1, 2, 3, \dots, q\}$ The number of codewords without using $\{1, 2, 3, \dots, q\}$ $= \left \bigcap_{i=1}^m \overline{A_i} \right $ $= S - \sum_i A_i + \sum_{i < j} A_i \cap A_j + \dots + (-1)^m \left \bigcap_{i=1}^m A_i \right $ $= (p)^k - \binom{m}{1} qp^{m-1} + \binom{m}{2} q^2 p^{m-2} + \dots + (-1)^m q^m$ $= \sum_{r=0}^m (-1)^m \binom{m}{r} q^r p^{m-r}$ It is known from part (iii) that the number of codewords without using $\{1, 2, 3, \dots, q\}$ is $(p - q)^m$. Hence we have $(p - q)^m = \sum_{r=0}^m \binom{m}{r} (-1)^r q^r p^{m-r}$
5(i)	For the k^{th} row, the path can choose to leave the row from the 1^{st} to the k^{th} triangle. Hence, its path can be represented as a sequence $\{a_1, a_2, \dots, a_{n-1}\}$, where $1 \leq a_i \leq i$. Conversely, for each such sequence, we can find a unique path. Hence, the number of paths equals to the number of such sequence, which is $(n-1)!$
5(ii)(a)	The length of the shortest path = $\begin{cases} 1 + 2(n-1) = 2n-1 & n \text{ is odd} \\ 2 + 2(n-2) = 2n-2 & n \text{ is even} \end{cases}$ $= 1 + 2 + 4 + 6 + \dots + 2(n-2) + (n-1)$ $= n + 2(1 + 2 + 3 + \dots + n-2)$ The length of the longest path = $n + 2 \frac{n-2}{2} (1 + n-2)$ $= n + (n-1)(n-2)$ $= n^2 - 2n + 2$
5(ii)(b)	When $n = 8$, the length of the longest path = 50, the length of the shortest path = 14.

	The total number of paths = $7! = 5040$ $5040 = (50 - 14 + 1)(136) + 8$ By Pigeonhole Principle, there exist at least one number m whereby there exists more than 136 paths of length m																
5(iii)(a)	The total number of triangles = n^2 The number of different colouring = $2^{n^2} - 1$ The number of different colouring that do not form a path = $2^{n^2} - 1 - (n - 1)!$																
5(iii)(b)	Case 1: The paths contain no perfect square other than 1 The total number of paths = $(8 - 2)!$ Case 2: The paths contains one perfect square other than 1 (the perfect square can be 4, 9, 16, 25, etc.) If the perfect square occurs in the k^{th} row, $2 \leq k \leq 7$ The number of paths = $\frac{(8 - 2)!}{(k - 1)}$ Hence, the total number of paths = $\sum_{k=2}^7 \frac{(8 - 2)!}{(k - 1)}$ The total number of paths $= (8 - 2)! + \sum_{k=2}^7 \frac{(8 - 2)!}{(k - 1)}$ $= 2484$																
6(i)	Euler Method: <table><tr><td>n</td><td>x_n</td><td>y_n</td><td>y_{n+1}</td></tr><tr><td>0</td><td>1</td><td>2</td><td>2.8</td></tr><tr><td>1</td><td>1.2</td><td>2.8</td><td>3.8</td></tr><tr><td>2</td><td>1.4</td><td>3.8</td><td>5.04</td></tr></table> Thus the approximate value is 5.04. Since the gradient values are positive and increasing, it is likely that the solution curve is concave upwards, and thus the approximate value is likely to be an underestimate.	n	x_n	y_n	y_{n+1}	0	1	2	2.8	1	1.2	2.8	3.8	2	1.4	3.8	5.04
n	x_n	y_n	y_{n+1}														
0	1	2	2.8														
1	1.2	2.8	3.8														
2	1.4	3.8	5.04														
6(ii)	Improved Euler Method : <table><tr><td>n</td><td>x_n</td><td>y_n</td><td>y_{n+1}</td></tr><tr><td>0</td><td>1.2</td><td>2.9</td><td>4.042</td></tr><tr><td>1</td><td>1.4</td><td>4.042</td><td>5.4792</td></tr></table> Thus the value of y at $x = 1.6$ is 5.4792 (exact).	n	x_n	y_n	y_{n+1}	0	1.2	2.9	4.042	1	1.4	4.042	5.4792				
n	x_n	y_n	y_{n+1}														
0	1.2	2.9	4.042														
1	1.4	4.042	5.4792														
6(iii)	$y = Ae^x - (x + 2)$ Substitute $x = 1, y = 2$ to obtain $A = 5e^{-1}$. Thus $y = 5e^{x-1} - (x + 2)$.																

	Substitute $x = 1.6$, $y = 5e^{0.6} - 3.6$ or 5.51 (3 s.f.)
6(iv)	Since 5.04 (Euler's method) < 5.48 (improved Euler's method) < 5.51 (actual value). Thus the improved Euler's method is a better method.
7	Suppose there are P kg of potassium permanganate in the lake at time t minutes. Then, $\frac{dP}{dt} = 900 \times 0.12 - \frac{1000P}{10^5 - 100t}$. Thus $\frac{dP}{dt} + \frac{10}{1000 - t}P = 108$. Integrating factor = $\exp\left(\int \frac{10}{1000 - t} dt\right)$ $= \exp(-10 \ln(1000 - t))$ $= (1000 - t)^{-10}$ The DE becomes $\frac{d}{dt}((1000 - t)^{-10} P) = 108(1000 - t)^{-10}$ $\Rightarrow (1000 - t)^{-10} P = \int 108(1000 - t)^{-10} dt$ $\Rightarrow (1000 - t)^{-10} P = 12(1000 - t)^{-9} + C$ $\Rightarrow P = 12(1000 - t) + C(1000 - t)^{10}$ Using initial given conditions: $P = 7000, t = 0$, $7000 = 12(1000) + C(1000)^{10}$ $\Rightarrow C = -5000(1000)^{-10}$ Thus $(1000 - t)^{-10} P = 12(1000 - t)^{-9} - 5(1000)^{-9}$ $\Rightarrow P = 12(1000 - t) - 5(1000)^{-9}(1000 - t)^{10}$ The required concentration (in kg per litre) is $\frac{P}{V} = \frac{12(1000 - t) - 5(1000)^{-9}(1000 - t)^{10}}{10^5 - 100t}$ $= \frac{12 - 5(1000)^{-9}(1000 - t)^9}{100} = \frac{3}{25} - \frac{1}{20} \left(1 - \frac{t}{1000}\right)^9$
8(i)	Since $b^2 - 4ac = 36 - 4 \times 1 \times 25 = -64 < 0$, therefore the system is underdamped.
8(ii)	The auxiliary roots are $x = \frac{-6 \pm \sqrt{-64}}{2} = -3 \pm 4i$. Thus the general solution is $x = e^{-3t} (A \cos 4t + B \sin 4t).$ At $t = 0, x = 0$, $0 = e^0 (A(1) + B(0)) \Rightarrow A = 0.$

	$\frac{dx}{dt} = e^{-3t} (-4A \sin 4t + 4B \cos 4t) - 3e^{-3t} (A \cos 4t + B \sin 4t).$ At $t = 0, \frac{dx}{dt} = 200, 200 = 4B - 3A \Rightarrow B = 50.$ Therefore, the particular solution is $x = 50e^{-3t} \sin 4t.$ Now, the amplitudes occur at $t = \frac{k\pi}{8},$ where k is an odd integer, are reducing as t increases. From table below: <table><tr><td>k</td><td>1</td><td>3</td><td>5</td></tr><tr><td>x</td><td>15.39</td><td>-1.46</td><td>0.138</td></tr></table> The first time the amplitude falls below 1cm is when $t = \frac{5\pi}{8}$ sec.	k	1	3	5	x	15.39	-1.46	0.138
k	1	3	5						
x	15.39	-1.46	0.138						
9(a) (i)	The constant r is the per capita growth rate of the toad population per hectare (growth rate per toad per hectare). The constant K is the carrying capacity of the toad population per hectare.								
9(a) (ii)	$\frac{dB}{dt} = rB \left(1 - \frac{B}{K} \right) = r \left(B - \frac{B^2}{K} \right)$ Since the graph of $\frac{dB}{dt}$ versus B is a concave downwards quadratic graph, the maximum rate occurs when $B = \frac{K}{2}.$  By a sketch of a phase diagram, we deduce that the population (per hectare) will increase and tend towards the carrying capacity $K.$								
9(b) (i)	The term represents the amount of toads being harvested per unit time (per hectare).								
9(b) (ii)	Let $\frac{dB}{dt} = rB \left(1 - \frac{B}{100} \right) - \frac{1}{r} = 0.$ Then $r^2 = \frac{100}{B(100 - B)}$ $\Rightarrow r^2 B^2 - 100r^2 B + 100 = 0$ $\Rightarrow B = 50 \pm \sqrt{2500 - \frac{100}{r^2}}.$ From this expression, we deduce there are two asymptotes $B = 0$ and $B = 100,$ and there is a vertex point $(r, B) = \left(\frac{1}{5}, 50 \right).$ The bifurcation diagram is shown								

as follows.



Note that the diagram is confined only to the first quadrant, with $r \neq 0$. The bifurcation value is $r = \frac{1}{5}$.

9(b)
(iii)

The range of r is $0 < r < \frac{1}{5}$